

DEPARTMENT OF MATHEMATICS
MAT111 INTRODUCTORY MATHEMATICS I
WEEK 10: LECTURE NOTE

Solving Exponential and Logarithmic equations

Example 1. *Solve the equation*

$$\log_7 49^{2x} = 6.$$

Solution:

Method 1: *By the definition of logarithm, we have*

$$\log_7 49^{2x} = 6$$

$$49^{2x} = 7^6$$

$$7^{2(2x)} = 7^6$$

$$4x = 6$$

$$x = \frac{3}{2}$$

Method 2: *By the property of logarithm,*

$$\log_7 49^{2x} = 6$$

$$2x \log_7 49 = 6$$

$$2x(2) = 6$$

$$4x = 6$$

$$x = \frac{6}{4} = \frac{3}{2}$$

Example 2. *Solve the equation*

$$\log_8(2x + 1) = \log_8\left(\frac{3}{5}\right).$$

Solution: *The domain of the equation is $(-\frac{1}{2}, \infty)$. Since a logarithmic function is one-to-one,*

$$\log_8(2x + 1) = \log_8\left(\frac{3}{5}\right)$$

$$2x + 1 = \frac{3}{5}$$

$$x = -\frac{1}{5}.$$

Example 3. *Solve the equation*

$$\ln(6 - 5x) = 2 \ln x.$$

Solution: *Domain of the equation is the interval $(0, \frac{6}{5})$. Note that in this domain,*

$$2 \ln x = \ln x^2.$$

Because y the natural logarithmic function is one-to-one, we have

$$\begin{aligned}\ln(6 - 5x) &= \ln x^2 \\ 6 - 5x &= x^2 \\ x^2 + 5x - 6 &= 0 \\ (x + 6)(x - 1) &= 0 \\ x = -6 \text{ or } x &= 1.\end{aligned}$$

Since -6 is not in the domain of the equation, 1 is the only solution of the equation.

Example 4. Solve the equation

$$\log_2(2x) - \log_2(x - 1) = \log_2 5.$$

Solution: Domain of the equation is the interval $(1, \infty)$. By using the properties of logarithmic function, we have:

$$\begin{aligned}\log_2(2x) - \log_2(x - 1) &= \log_2 5 \\ \log_2\left(\frac{2x}{x - 1}\right) &= \log_2 5 \\ \frac{2x}{x - 1} &= 5 \\ x &= \frac{5}{3}.\end{aligned}$$

Example 5. Use the properties of logarithms to expand the following expressions as a sum, difference, and/or constant multiple of logarithms.

$$(a) \log_3(9x^4\sqrt{y}) \quad (b) \log_5\left[\frac{5}{2x^2(1 - y^3)}\right] \quad (c) \ln\left(\frac{\sqrt{3x - 5}}{7}\right)$$

Solution:

(a) By using the properties of logarithmic function, we have

$$\log_3(9x^4\sqrt{y}) = \log_3 9 + \log_3 x^4 + \log_3 \sqrt{y} = 2 + 4\log_3 x + \frac{1}{2}\log_3 y$$

(b) By using the properties of logarithmic function, we have

$$\begin{aligned}\log_5\left[\frac{5}{2x^2(1 - y^3)}\right] &= \log_5 5 - \log_5[2x^2(1 - y^3)] \\ &= 1 - [\log_5 2 + \log_5 x^2 + \log_5(1 - y^3)] \\ &= 1 - \log_5 2 - 2\log_5 x - \log_5(1 - y^3)\end{aligned}$$

(c) By using the properties of logarithmic function, we have

$$\ln\left(\frac{\sqrt{3x - 5}}{7}\right) = \ln \sqrt{3x - 5} - \ln 7 = \frac{1}{2}\ln(3x - 5) - \ln 7$$

Example 6. Use properties of logarithms to solve the following equations:

$$(a) \ e^x = 66 \qquad (b) \ 4^x = 100 \qquad (c) \ 2^{x+5} = 256 \qquad (d) \ 5(2^{3-x}) - 13 = 100$$

Solution:

(a) By using the properties of natural logarithmic function, we have

$$\begin{aligned} e^x &= 66 \\ \ln e^x &= \ln 66 \\ x &= \ln 66 = 4.19 \end{aligned}$$

(b) By using the properties of natural logarithmic function, we have

$$\begin{aligned} 4^x &= 100 \\ \ln 4^x &= \ln 100 \\ x \ln 4 &= \ln 100 \\ x &= \frac{\ln 100}{\ln 4} = 3.32 \end{aligned}$$

(c) By using the properties of natural logarithmic function, we have

$$\begin{aligned} 2^{x+5} &= 256 \\ \ln(2^{x+5}) &= \ln 256 \\ (x+5) \ln 2 &= \ln 256 \\ x &= \frac{\ln 256}{\ln 2} - 5 = 8 - 5 = 3. \end{aligned}$$

(d) By using the properties of natural logarithmic function, we have

$$\begin{aligned} 5(2^{3-x}) - 13 &= 100 \\ 5(2^{3-x}) &= 113 \\ 2^{3-x} &= \frac{113}{5} \\ (3-x) \ln 2 &= \ln \left(\frac{113}{5} \right) = \ln (22.6) \\ x &= -\frac{\ln (22.6)}{\ln 2} + 3 = -1.5 \end{aligned}$$

Example 7. Use properties of logarithms to condense the following expressions.

$$(a) \ \frac{1}{2} \log_{10} x + 3 \log_{10}(x+1) \qquad (b) \ 2 \ln(x+2) - \ln(5x-1)$$

Solution:

(a) By using the properties of logarithm, we have

$$\frac{1}{2} \log_{10} x + 3 \log_{10}(x+1) = \log_{10} \sqrt{x} + \log_{10}(x+1)^3 = \log_{10}(\sqrt{x}(x+1)^3)$$

(b) By using the properties of natural logarithm, we have

$$2 \ln(x+2) - \ln(5x-1) = \ln(x+2)^2 - \ln(5x-1) = \ln \frac{(x+2)^2}{5x-1}$$

EXERCISES

1. Use properties of logarithmic functions and exponential functions to solve the following equations.

(a) $\log_3 3^{-5} + \log_5 5 = x$

(b) $x = \log_4 25$

(c) $\ln 4 + \ln (2 - 3x) = \ln (3 - 2x)$

(d) $\log_3 (2x + 5)^2 = 4$

(e) $3^{x-2} = \frac{1}{81}$

(f) $e^{x^2-3x} = e^{x-2}$

2. Use properties of logarithms to expand (or to condense) the following expressions.

(a) $\ln\left(\frac{e^2}{\sqrt{2x^2(3+y)}}\right)$

(b) $\frac{1}{3}[\log_2 x - \log_2 (2x + 5)]$

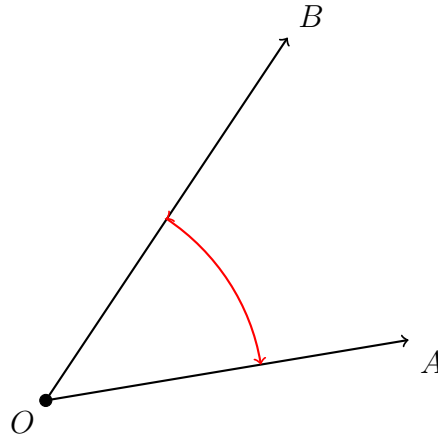
(c) $\frac{1}{2}[\ln 64 - \ln (x^2 + 3)]$

TRIGONOMETRIC FUNCTIONS

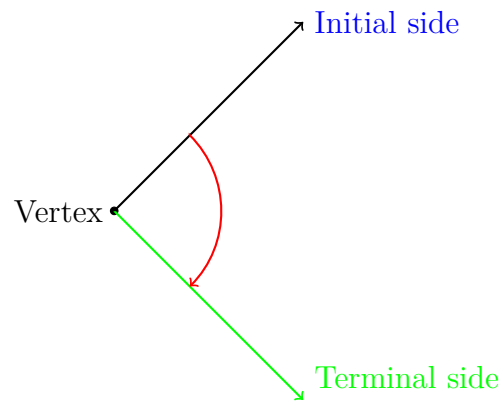
ANGLES AND THEIR MEASURES

THE NOTION OF ANGLES

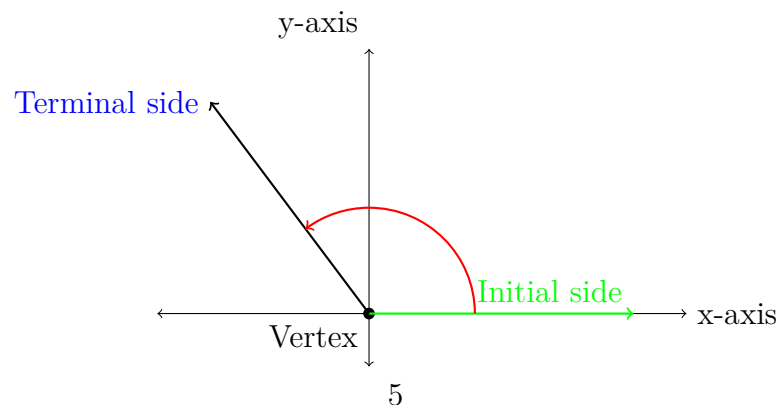
In geometry, an angle can be defined as the figure formed by two rays meeting at a common end point. An angle is represented by the symbol $\angle$. Here, the angle below is $\angle AOB$.



An angle is determined by rotating a ray (half-line) about its endpoint. The starting position of the ray is the initial side of the angle, and the position after rotation is the terminal side. The endpoint of the ray is the vertex of the angle.



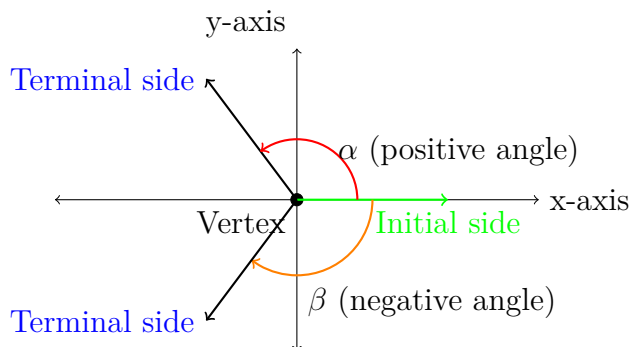
THE STANDARD POSITION: An angle is said to be in standard position if its vertex is located at the origin, and its initial side extends along the positive x-axis.



From the standard position:

- Positive angles are generated by **counterclockwise rotation**.
- Negative angles are generated by **clockwise rotation**.

Labeling Angles: Angles are labeled with Greek letters such as α , β , and θ , as well as uppercase letters such as A , B , and C .



From the standard position, angle denoted by α is positive; while angle β is negative.

DEGREE MEASURE

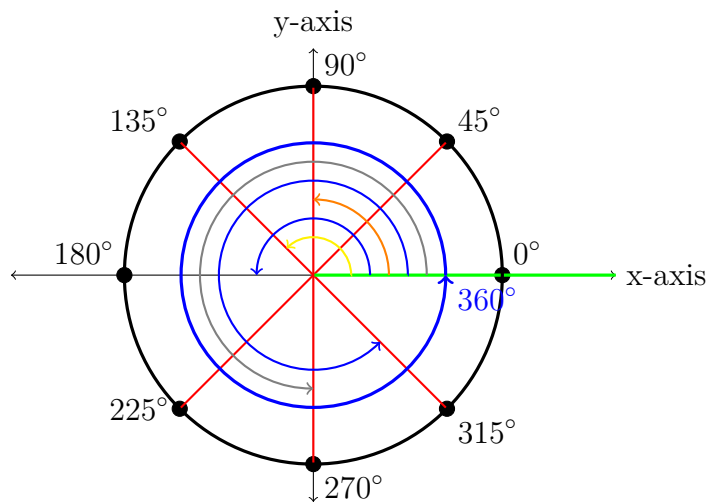
The measure of an angle is determined by **the amount of rotation from the initial side to the terminal side**. One way to measure an angle is in terms of **degrees**, denoted by $^\circ$. A measure of **one degree** (1°) is equivalent to a rotation of $\frac{1}{360}$ of a complete revolution. Thus,

1. a complete revolution is 360° ;
2. half a revolution is 180° , called the **straight angle**.
3. quarter of a revolution is 90° , called the **right angle**.
4. one-third of a revolution is 120° .
5. two-thirds of a revolution is 240° .
6. three-fourths of a revolution is 270° .
7. one-sixth of a revolution is 60° .

Two angles are said to be **coterminal** when they have the same initial and terminal sides. For instance

1. 0° and 360° are coterminal.
2. -90° and 270° are coterminal.
3. 45° and 405° are coterminal.

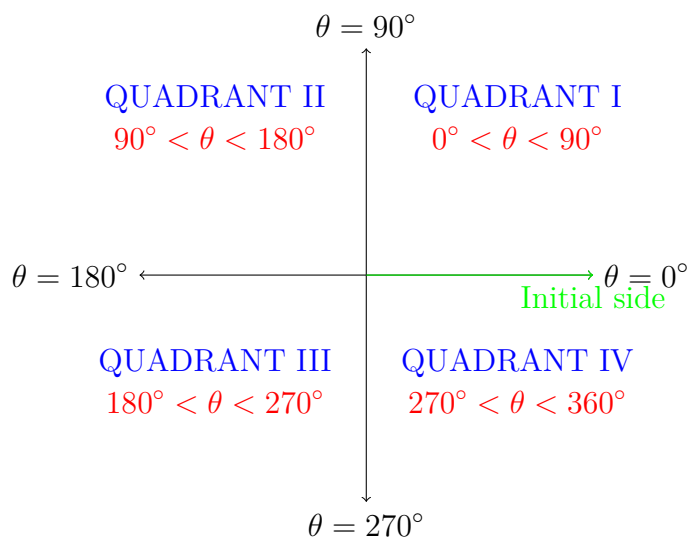
To measure angles, it is convenient to mark degrees on the circumference of a circle.



Experiment for Students: Draw a circle of radius 10cm . Use your protractor to draw and measure angles.

1. Putting initial side on standard position, draw angles: 30° , 60° , 90° , 120° , 150° , 180° , 210° , 240° , 270° , 300° , 330° .
2. Putting initial side on standard position, draw angles: -45° , -90° , -135° , -180° , -225° , -270° , -315° , -360° .
3. Putting initial side on standard position, draw angles: -30° , -60° , -120° , -150° , -210° , -240° , -300° , -330° .
4. Putting initial side on standard position, draw angles: 36° , 72° , 108° , 144° , 216° , 252° , 288° , 324° .
5. Putting initial side on standard position, draw angles: 10° , 20° , 40° , 50° , 70° , 80° , 100° , 110° , 130° , 140° , 160° , 170° , 190° , 200° , 220° , 230° , 250° , 260° , 280° , 290° , 310° , 320° , 340° , 350° , 450° , 540° , 720° .

Summary of angles whose terminal sides fall in different quadrants:



TYPES OF ANGLES

There are various types of angles based on their degree measures. The types are:

1. Acute angle: $0^\circ < \theta < 90^\circ$
2. Right angle (or quarter revolution): $\theta = 90^\circ$
3. Obtuse angle: $90^\circ < \theta < 180^\circ$
4. Straight angle (or half revolution) : $\theta = 180^\circ$
5. Reflex angle: $180^\circ < \theta < 360^\circ$

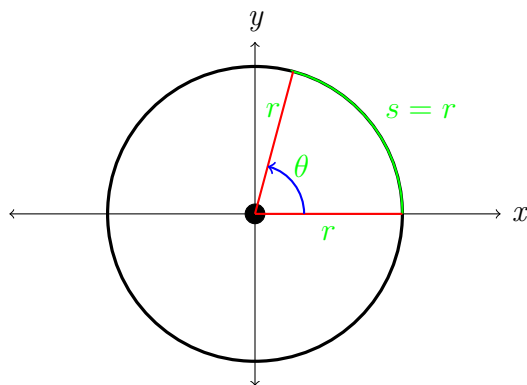
Definition 8 (Complementary and Supplementary Angles). 1. If two positive angles add up to measure 90° , then they are known as **complementary angles**.

2. When two positive angles add up to 180° , then they are known as **supplementary angles**.

RADIAN MEASURE

Second way to measure angles is in **radians**.

Definition 9. One radian (rad) is the measure of a central angle θ that intercepts an arc equal in length to the radius of the circle. Algebraically this means that $\theta = \frac{s}{r}$, where s is θ measured in radians.



Note the following:

1. The circumference of a circle divide by its diameter is a constant number π , which irrational number, approximated as 3.14 or $\frac{22}{7}$ for practical purposes.
2. The circumference of a circle of radius r is $2\pi r$ units. It follows that a central angle of **one full revolution** (counterclockwise) corresponds to an arc length of

$$s = 2\pi r \simeq 6.28r.$$

So that one full revolution is 2π radians.

3. Conversions Between **Degrees** and **Radians**:

(a) To convert degrees to radians, multiply degrees by $\frac{\pi rad}{180^\circ}$.

(b) To convert radians to degrees, multiply radians by $\frac{180^\circ}{\pi rad}$.

To apply these two conversion rules, use the basic relationship

$$\pi \text{ rad} = 180^\circ.$$

Example 10 (Converting from Degrees to Radians). 1. $90^\circ = 90^\circ \times \frac{\pi \text{ rad}}{180^\circ} = \frac{\pi}{2} \text{ rad}$

$$2. -30^\circ = -30^\circ \times \frac{\pi \text{ rad}}{180^\circ} = -\frac{\pi}{6} \text{ rad}$$

$$3. 150^\circ = 150^\circ \times \frac{\pi \text{ rad}}{180^\circ} = \frac{5\pi}{6} \text{ rad}$$

$$4. -300^\circ = -300^\circ \times \frac{\pi \text{ rad}}{180^\circ} = -\frac{5\pi}{3} \text{ rad}$$

$$5. 540^\circ = 540^\circ \times \frac{\pi \text{ rad}}{180^\circ} = 3\pi \text{ rad}$$

Example 11 (Converting from Radians to Degrees). 1. $3 \text{ rad} = 3 \text{ rad} \times \frac{180^\circ}{\pi \text{ rad}} = \left(\frac{540}{\pi}\right)^\circ \approx 172^\circ$

$$2. \frac{\pi}{6} \text{ rad} = \frac{\pi}{6} \text{ rad} \times \frac{180^\circ}{\pi \text{ rad}} = 30^\circ$$

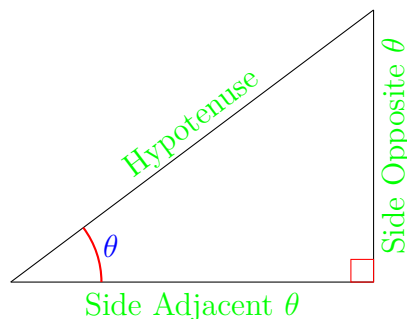
$$3. \frac{5\pi}{9} \text{ rad} = \frac{5\pi}{9} \text{ rad} \times \frac{180^\circ}{\pi \text{ rad}} = 100^\circ$$

$$4. 5\pi \text{ rad} = 5\pi \text{ rad} \times \frac{180^\circ}{\pi \text{ rad}} = 900^\circ$$

RIGHT TRIANGLE TRIGONOMETRY

THE SIX TRIGONOMETRIC FUNCTIONS

Consider a right triangle with one acute angle labeled θ , as shown in the figure below. Relative to the angle θ , the three sides of the triangle are the **hypotenuse**, the **opposite side** (the side opposite the angle), and the **adjacent side** (the side adjacent to the angle).



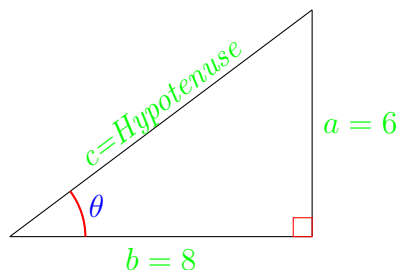
Using the lengths of these three sides, we can form **six ratios**.

Definition 12. *These six ratios define the six trigonometric functions sine, cosecant, cosine, secant, tangent and cotangent of the acute angle θ , denoted by $\sin \theta$, $\csc \theta$, $\cos \theta$, $\sec \theta$, $\tan \theta$ and $\cot \theta$, respectively; and are given by*

1. $\sin \theta = \frac{\text{the length of the side opposite } \theta}{\text{the length of hypotenuse}}$
2. $\csc \theta = \frac{\text{the length of hypotenuse}}{\text{the length of the side opposite } \theta}$
3. $\cos \theta = \frac{\text{the length of the side adjacent } \theta}{\text{the length of hypotenuse}}$
4. $\sec \theta = \frac{\text{the length of hypotenuse}}{\text{the length of the side adjacent } \theta}$
5. $\tan \theta = \frac{\text{the length of the side opposite } \theta}{\text{the length of the side adjacent } \theta}$
6. $\cot \theta = \frac{\text{the length of the side adjacent } \theta}{\text{the length of the side opposite } \theta}$

Note that the functions in (2), (4) and (6) are the reciprocals of the functions in (1), (3) and (5), respectively.

Example 13. *Use the triangle in the Figure below to find the exact values of the six trigonometric functions of θ .*



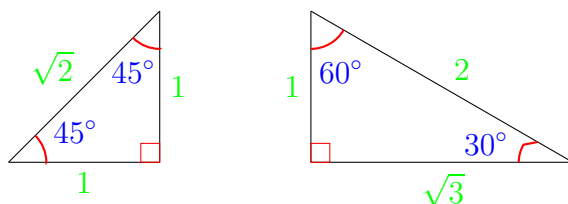
Solution. By the Pythagorean Theorem, $c^2 = a^2 + b^2 = 6^2 + 8^2 = 100$; so that $c = 10$.

$$\begin{array}{lll}
 (1) \sin \theta = \frac{a}{c} = \frac{3}{5} & (3) \cos \theta = \frac{b}{c} = \frac{4}{5} & (5) \tan \theta = \frac{a}{b} = \frac{3}{4} \\
 (2) \csc \theta = \frac{c}{a} = \frac{5}{3} & (4) \sec \theta = \frac{c}{b} = \frac{5}{4} & (6) \cot \theta = \frac{b}{a} = \frac{4}{3}
 \end{array}$$

TRIGONOMETRIC FUNCTIONS OF SPECIAL ANGLES

Special acute angles: 30° , 45° , 60° .

Derive the trigonometric ratios of 30° , 45° , and 60° from the 30-60-90 and 45-45-90 special triangles in the figure below.

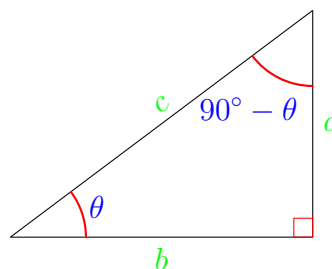


θ in degrees	θ in radians	$\sin \theta$	$\cos \theta$	$\csc \theta$	$\sec \theta$	$\tan \theta$	$\cot \theta$
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	2	$\frac{2}{\sqrt{3}}$	$\frac{1}{\sqrt{3}}$	$\sqrt{3}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	$\sqrt{2}$	$\sqrt{2}$	1	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\frac{2}{\sqrt{3}}$	2	$\sqrt{3}$	$\frac{1}{\sqrt{3}}$

TRIGONOMETRIC IDENTITIES

In trigonometry, a great deal of time is spent studying relationships between trigonometric functions (identities). Consider an acute angle θ of the right triangle.

1. $\sin \theta = \frac{a}{c} = \cos(90^\circ - \theta)$
2. $\cos \theta = \frac{b}{c} = \sin(90^\circ - \theta)$
3. $\tan \theta = \frac{a}{b} = \cot(90^\circ - \theta)$
4. $\cot \theta = \frac{b}{a} = \tan(90^\circ - \theta)$
5. $\csc \theta = \frac{c}{a} = \sec(90^\circ - \theta)$
6. $\sec \theta = \frac{c}{b} = \csc(90^\circ - \theta)$



Therefore, cofunctions of complementary angles are equal.

NB. When evaluating a trigonometric function with a calculator, you need to set the calculator to the desired mode of measurement (degrees or radians).

Notation: $\sin^2 \theta$ denotes $(\sin \theta)^2$; $\cos^2 \theta$ denotes $(\cos \theta)^2$, and so on.

Proposition 14 (FUNDAMENTAL TRIGONOMETRIC IDENTITIES). *I. Reciprocal Identities*

- $\sin \theta = \frac{1}{\csc \theta}$
- $\csc \theta = \frac{1}{\sin \theta}$
- $\cos \theta = \frac{1}{\sec \theta}$
- $\sec \theta = \frac{1}{\cos \theta}$
- $\tan \theta = \frac{1}{\cot \theta}$
- $\cot \theta = \frac{1}{\tan \theta}$

II. Quotient Identities

- $\tan \theta = \frac{\sin \theta}{\cos \theta}$
- $\cot \theta = \frac{\cos \theta}{\sin \theta}$

III. Pythagorean Identities

- $\sin^2 \theta + \cos^2 \theta = 1$
- $1 + \tan^2 \theta = \sec^2 \theta$
- $1 + \cot^2 \theta = \csc^2 \theta$

The above trigonometric identities are easily proved for acute angle θ of a right triangle. It is left as an exercise.

Example 15. Let θ be an acute angle such that $\sin \theta = \frac{5}{13}$. Find the values of $\cos \theta$, $\tan \theta$, $\cot \theta$, $\csc \theta$ and $\sec \theta$ using trigonometric identities.

Solution: It follows from Pythagorean Identities

$$1 = \sin^2 \theta + \cos^2 \theta = \frac{25}{169} + \cos^2 \theta;$$

so that

$$\cos^2 \theta = 1 - \frac{25}{169} = \frac{144}{169} \Rightarrow \cos \theta = \frac{12}{13}.$$

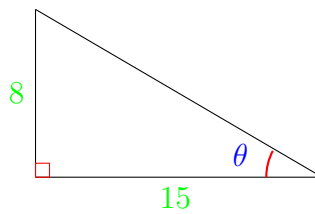
It follows from trigonometric identities

$$\otimes \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{5}{12}, \quad \otimes \cot \theta = \frac{\cos \theta}{\sin \theta} = \frac{12}{5},$$

$$\otimes \csc \theta = \frac{1}{\sin \theta} = \frac{13}{5}, \quad \otimes \sec \theta = \frac{1}{\cos \theta} = \frac{13}{12}.$$

EXERCISES

1. Determine two coterminal angles in degree measure (one positive and one negative) for the angles $\theta = 300^\circ$ and $\alpha = -740^\circ$.
2. Find (if possible) the complement and supplement of the angle $\theta = 24^\circ$.
3. Rewrite angle $\theta = 30^\circ$ in radian measure as a multiple of π .
4. Rewrite angle $\theta = \frac{3\pi}{2}$ rad in degree measure.
5. Find (if possible) the complement and supplement of the angle $\theta = \frac{\pi}{3}$.
6. Find the length of the arc on a circle of radius $r = 14$ inches intercepted by a central angle $\theta = 180^\circ$.
7. Linear Speed: A satellite in a circular orbit 1250 kilometers above Earth makes one complete revolution every 110 minutes. What is its linear speed? Assume that Earth is a sphere of radius 6400 kilometers.
8. Mechanical Engineering: The circular blade on a saw has a diameter of 7.5 inches and rotates at 2400 revolutions per minute.
9. Find the angular speed in radians per second.
10. Find the linear speed of the saw teeth (in feet per second) as they contact the wood being cut.
11. Find the exact values of the six trigonometric functions of the angle θ shown in the figure below.



12. Use a calculator to evaluate $\sin 10^\circ$. Round your answer to four decimal places.
13. Use the given function value(s) and the trigonometric identities to find the indicated trigonometric functions.

a. $\csc \theta = 3$, $\sec \theta = \frac{3\sqrt{2}}{4}$. Find

(a) $\sin \theta$ (b) $\cos \theta$ (c) $\tan \theta$ (d) $\sec(90^\circ - \theta)$

b. $\cos \alpha = \frac{1}{3}$. Find

(a) $\sec \alpha$ (b) $\sin \alpha$ (c) $\cot \alpha$ (d) $\sin(90^\circ - \alpha)$

14. For $0 < \theta < \frac{\pi}{2}$, use trigonometric identities to transform one side of the following equation into the other.

$$\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta} = \csc \theta \sec \theta.$$

15. **GEOMETRY** A six-foot person walks from the base of a streetlight directly toward the tip of the shadow cast by the streetlight. When the person is 16 feet from the streetlight and 5 feet from the tip of the streetlight's shadow, the person's shadow starts to appear beyond the streetlight's shadow.
- (a) Draw a right triangle that gives a visual representation of the problem. Show the known quantities and use a variable to indicate the height of the streetlight.
- (b) Use a trigonometric function to write an equation involving the unknown quantity.
- (c) What is the height of the streetlight?
16. **ANGLE OF ELEVATION.** You are skiing down a mountain with a vertical height of 1500 feet. The distance from the top of the mountain to the base is 3000 feet. What is the angle of elevation from the base to the top of the mountain?
17. **BIOLOGY** A biologist wants to know the width w of a river (see figure) in order to properly set instruments for studying the pollutants in the water. From point A , the biologist walks upstream 100 feet and sights to point C . From this sighting, it is determined that $\theta = 58^\circ$. How wide is the river? Verify your result numerically.

